Joint convexity of the Umegaki quantum relative entropy

A conventional-mathematics proof, in exact correspondence with the Lean formalization `qRelativeEnt_joint_convexity` (PhysLean)

Prepared for review — PhysLean contribution

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We give a self-contained, conventional-mathematics proof that the Umegaki quantum relative entropy is jointly convex. The argument is presented exactly as it is formalized in Lean 4 in the PhysLean library (theorem `qRelativeEnt_joint_convexity`, commit `720c9fff`): every step below corresponds to an identifiable step in the formal proof, and §7 gives the full paper-to-Lean dictionary. The proof route is the Frank–Lieb / Leditzky–Rouzé–Datta variational argument for the sandwiched Rényi trace functional, taken to the limit $\alpha \to 1^+$; it deliberately avoids differentiability at $\alpha = 1$, using instead a one-sided convergent majorant (Lemma E, the one genuinely new ingredient) that requires only continuity plus two elementary scalar bounds. The purpose of this paper is to let a reviewer confirm the *mathematical* correctness of the contribution in ordinary language, independently of reading Lean, while §7 makes the correspondence to the machine-checked proof precise.

This note renders the Lean proof of `qRelativeEnt_joint_convexity` (`QuantumInfo/Entropy/DPI.lean`, line 1534 as of commit `720c9fff`) as a mathematician would write it on paper. The structure of the argument below follows the Lean proof step for step; Lean identifiers are given in `code font` at the point where they are used. The main theorem delegates its constituent steps to a handful of supporting lemmas — the public `sandwichedRelRentropy_tendsto_qRelativeEnt` and the private `mix_M_eq_weighted_sum`, `ker_mix_le`, `sandwichedTraceFunctional_mix_le` — that sit immediately above it in `DPI.lean`; §7 gives the full correspondence. Nothing is included that is not in the formal proof, and the two pre-existing ingredients it relies on (joint convexity of the trace functional, and continuity in α) are stated as propositions with their own provenance.

1. Setting and conventions

Throughout, d is a finite index set and states live on $\mathbb{C}^d$.

- A **state** (`MState d`, `QuantumInfo/States/Mixed/MState.lean:63`) is a Hermitian matrix $\rho \geq 0$ with $\operatorname{Tr} \rho = 1$.
- For $A \geq 0$ and $r \in \mathbb{R}$, the power A^r is defined by functional calculus with $0^r := 0$ for $r \neq 0$ (`HermitianMat.rpow`, `QuantumInfo/ForMathlib/HermitianMat/Rpow.lean:45`); negative

powers are therefore Moore–Penrose-style powers on the support. Likewise $\log A$ is taken on the support.

- We write $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ for the support (= range) inclusion of PSD matrices; in Lean this is the kernel inclusion `σ.M.ker ≤ ρ.M.ker`.
- Entropies take values in $[0,\infty]$ ($\mathbb{R}_{\geq 0\infty}$), with the conventions $a \cdot \infty = \infty$ for $a \neq 0$ and $0 \cdot \infty = 0$.
- p ranges over `Prob` $= \{p \in \mathbb{R} \mid 0 \leq p \leq 1\}$ (`QuantumInfo/ClassicalInfo/Prob.lean:36`). The mixture notation `p [ρ₁ ↔ ρ₂]` (`Mixable.mix`, `Prob.lean:267`) denotes the state whose matrix is $p\rho_1 + (1-p)\rho_2$. In the theorem statement, `1 - p` on the right-hand side is truncated subtraction in $\mathbb{R}_{\geq 0\infty}$, which equals the real $1-p$ since $p \leq 1$.
- log is the natural logarithm.

2. Definitions

Definition 1 (Umegaki relative entropy; `qRelativeEnt`, notation D, `Relative.lean:1497`).

$$D(\rho \,\|\, \sigma) \;=\; \begin{cases} \operatorname{Tr}\big[\rho\,(\log\rho - \log\sigma)\big] & \text{if } \operatorname{supp}\rho \subseteq \operatorname{supp}\sigma, \\ +\infty & \text{otherwise.} \end{cases}$$

(In Lean, **D**$(\rho\|\sigma)$ is *defined* as `D_1`$(\rho\|\sigma)$, the $\alpha = 1$ member of the next definition; the identification with the trace formula under the support condition is `qRelativeEnt_ker`, `Relative.lean:1782`.)

Definition 2 (sandwiched Rényi relative entropy; `SandwichedRelRentropy`, notation `D_`α, `Relative.lean:1480`). For $\alpha > 0$, set $\gamma := \frac{1-\alpha}{2\alpha}$. If $\operatorname{supp}\rho \not\subseteq \operatorname{supp}\sigma$, then $\widetilde{D}_\alpha(\rho\|\sigma) := +\infty$. Otherwise

$$\widetilde{D}_\alpha(\rho \,\|\, \sigma) \;:=\; \begin{cases} \dfrac{1}{\alpha-1}\,\log\operatorname{Tr}\big[(\sigma^\gamma\rho\,\sigma^\gamma)^\alpha\big] & \alpha \neq 1, \\ \operatorname{Tr}\big[\rho\,(\log\rho - \log\sigma)\big] & \alpha = 1, \end{cases}$$

a nonnegative quantity (`sandwichedRelRentropy_nonneg`). (For $\alpha \leq 0$ the Lean definition returns the junk value 0; only $\alpha \geq 1$ is used here.)

Definition 3 (sandwiched trace functional; `sandwichedTraceFunctional`, notation `Q_`α, `DPI.lean:66`). For $\alpha \in \mathbb{R}$ and $\gamma = \frac{1-\alpha}{2\alpha}$,

$$\widetilde{Q}_\alpha(\rho \,\|\, \sigma) \;:=\; \operatorname{Tr}\big[(\sigma^\gamma\rho\,\sigma^\gamma)^\alpha\big] \;\in\; \mathbb{R}.$$

For $0 < \alpha \neq 1$ and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ one has (`sandwichedRelRentropy_eq_log_traceFunctional`, `DPI.lean:78`)

$$\widetilde{D}_\alpha(\rho\|\sigma) = \Big[\tfrac{1}{\alpha-1}\log\widetilde{Q}_\alpha(\rho\|\sigma)\Big]^+ \quad \text{(as an element of } [0,\infty],\ \text{via ofReal).}$$

3. Theorem

Theorem (`qRelativeEnt_joint_convexity`, `DPI.lean:1522`). For all states $\rho_1, \rho_2, \sigma_1, \sigma_2$ on $\mathbb{C}^d$ and all $p \in [0,1]$, writing $\bar\rho := p\rho_1 + (1-p)\rho_2$ and $\bar\sigma := p\sigma_1 + (1-p)\sigma_2$,

$$D(\bar\rho \,\|\, \bar\sigma) \;\leq\; p\,D(\rho_1\|\sigma_1) \;+\; (1-p)\,D(\rho_2\|\sigma_2)$$

as an inequality in $[0,\infty]$.

```
theorem qRelativeEnt_joint_convexity :
    ∀ (ρ₁ ρ₂ σ₁ σ₂ : MState d), ∀ (p : Prob),
      𝐃(p [ρ₁ ↔ ρ₂]‖p [σ₁ ↔ σ₂]) ≤ p * 𝐃(ρ₁‖σ₁) + (1 - p) * 𝐃(ρ₂‖σ₂)
```

4. Ingredients

The proof consumes two substantial pre-existing results (A, B), one elementary matrix lemma (C), and two scalar inequalities from Mathlib (D). Only Lemma E and the assembly in §5 are new in this contribution.

Proposition A (joint convexity of $\widetilde{Q}_\alpha$, $\alpha > 1$; `sandwichedTraceFunctional_jointly_convex`, `DPI.lean:568`, pre-existing). Let $\alpha > 1$, let ι be a finite index set, let $w_i \geq 0$ with $\sum_i w_i = 1$, and let (ρ_i), (σ_i) be states with $\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i$ for every i. If $\bar\rho, \bar\sigma$ are states with $\bar\rho = \sum_i w_i\rho_i$ and $\bar\sigma = \sum_i w_i\sigma_i$ as matrices, then

$$\widetilde{Q}_\alpha(\bar\rho \,\|\, \bar\sigma) \;\leq\; \sum_i w_i\, \widetilde{Q}_\alpha(\rho_i\|\sigma_i).$$

Proof route (already formalized; sketch only). For $\alpha > 1$ and $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ one has the variational formula (`traceFunctional_eq_iSup_f_alpha`, `DPI.lean:434`)

$$\widetilde{Q}_\alpha(\rho\|\sigma) = \sup_{H \geq 0} f_\alpha(H, \rho, \sigma), \qquad f_\alpha(H,\rho,\sigma) = \alpha \operatorname{Tr}[\rho H] - (\alpha - 1)\operatorname{Tr}\!\left[(\sigma^{-\gamma} H \sigma^{-\gamma})^{\alpha/(\alpha-1)}\right]$$

(`f_alpha`, `DPI.lean:174`), the supremum being attained at $\hat H = \sigma^\gamma(\sigma^\gamma\rho\,\sigma^\gamma)^{\alpha-1}\sigma^\gamma$. For fixed $H \geq 0$, f_α is linear in ρ and convex in σ: with $s := -\gamma = \frac{\alpha-1}{2\alpha} \in (0, \frac{1}{2})$ and $q := \frac{\alpha}{\alpha-1} > 1$, the map $\sigma \mapsto \operatorname{Tr}[(\sigma^s H \sigma^s)^q]$ is concave on the PSD cone (`HermitianMat.trace_conj_rpow_concave`, `QuantumInfo/ForMathlib/HermitianMat/LiebConcavity.lean:494`, itself derived from the Lieb–Ando trace theorems formalized in `QuantumInfo/ForMathlib/HayataGroup/TraceInequality/LiebAndoTrace.l` and it enters f_α with the negative coefficient $-(\alpha - 1)$. Hence each $f_\alpha(H, \cdot, \cdot)$ is jointly convex (`f_alpha_jointly_convex`, `DPI.lean:494`), and a pointwise supremum of jointly convex functions is jointly convex (`iSup_f_alpha_jointly_convex`, `DPI.lean:527`). This is the Frank–Lieb variational argument in the form used by Leditzky–Rouzé–Datta; see the literature note. ☐

Proposition B (continuity in α; `sandwichedRelRentropy.continuousOn`, `Relative.lean:1769`, pre-existing). For fixed states ρ, σ, the map $\alpha \mapsto \widetilde{D}_\alpha(\rho\|\sigma)$ is continuous on $(0, \infty)$ (with respect to the order topology on $[0, \infty]$). The consequence used below is:

$$\widetilde{D}_\alpha(\rho\|\sigma) \longrightarrow \widetilde{D}_1(\rho\|\sigma) = D(\rho\|\sigma) \qquad (\alpha \to 1^+).$$

Note that since $\widetilde{D}_1 := D$ *by definition*, the $\alpha \to 1$ convergence of the sandwiched Rényi entropy to the Umegaki entropy is precisely the content of this continuity theorem; it is a genuine analytic fact proved earlier in the library, not a definitional triviality.

Lemma C (kernels of weighted sums; `HermitianMat.ker_weighted_sum_le`, `DPI.lean:545`). If $w_i \geq 0$, all $\rho_i, \sigma_i \geq 0$, and $\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i$ for all i, then $\operatorname{supp}(\sum_i w_i\rho_i) \subseteq \operatorname{supp}(\sum_i w_i\sigma_i)$.

Scalar facts D (Mathlib). (i) $\log x \leq x - 1$ for $x > 0$ (`Real.log_le_sub_one_of_pos`). (ii) $|e^x - 1 - x| \leq x^2$ for $|x| \leq 1$ (`Real.abs_exp_sub_one_sub_id_le`); in particular $e^x - 1 \leq x + x^2$ there.

Also used: strict positivity $\widetilde{Q}_\alpha(\rho\|\sigma) > 0$ under the support condition (`sandwichedTraceFunctional_pos`, `DPI.lean:100`).

5. The key new lemma: a convergent majorant for the difference quotient

Lemma E (`sandwichedTraceFunctional_sub_one_div_eventually_le`, `DPI.lean:1432`, new). Let ρ, σ be states with $\operatorname{supp}\rho \subseteq \operatorname{supp}\sigma$ (so $D(\rho\|\sigma) < \infty$). Then there is a function $u : \mathbb{R} \to \mathbb{R}$ with

$$u(\alpha) \xrightarrow[\alpha\to 1^+]{} D(\rho\|\sigma) \qquad \text{and} \qquad \frac{\widetilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \le u(\alpha) \quad \text{for all } \alpha > 1 \text{ sufficiently close to 1.}$$

Proof. For $\alpha > 1$ put

$$r(\alpha) := \frac{\log \widetilde{Q}_\alpha(\rho\|\sigma)}{\alpha - 1}, \qquad x(\alpha) := (\alpha - 1)\, r(\alpha) = \log \widetilde{Q}_\alpha(\rho\|\sigma),$$

and define $u(\alpha) := r(\alpha) + x(\alpha)\, r(\alpha) = r(\alpha) + (\alpha - 1) r(\alpha)^2$.

1. Under the support condition, $r(\alpha) \ge 0$ and $\widetilde{D}_\alpha(\rho\|\sigma) = [r(\alpha)]^+$ (Definition 3 with `sandwichedRelRentropy_nonneg`), and $D(\rho\|\sigma) \neq \infty$ (Lean: `h_ne`, from `qRelativeEnt_ne_top_iff.mpr hker`; `Relative.lean:1789`). By Proposition B, packaged as `sandwichedRelRentropy_tendsto_qRelative` (`DPI.lean:1423`), $r(\alpha) \to D(\rho\|\sigma)$ (as real numbers) as $\alpha \to 1^+$ (Lean: `h_r_tendsto`, passing through `ENNReal.tendsto_toReal`).
2. Hence $x(\alpha) = (\alpha - 1) r(\alpha) \to 0$, so eventually $|x(\alpha)| \le 1$ (Lean: `h_eps`, `h_small`).
3. $\widetilde{Q}_\alpha(\rho\|\sigma) > 0$ (positivity above), so $\widetilde{Q}_\alpha(\rho\|\sigma) = e^{x(\alpha)}$. By D(ii), for the eventual α of step 2, $\widetilde{Q}_\alpha - 1 \le x(\alpha) + x(\alpha)^2$; dividing by $\alpha - 1 > 0$,
$$\frac{\widetilde{Q}_\alpha(\rho\|\sigma) - 1}{\alpha - 1} \;\le\; r(\alpha) + x(\alpha)\, r(\alpha) \;=\; u(\alpha).$$
4. By steps 1–2, $u(\alpha) \to D(\rho\|\sigma) + 0 \cdot D(\rho\|\sigma) = D(\rho\|\sigma)$. ∎

Heuristic remark (not part of the formal proof). Since $\widetilde{Q}_1(\rho\|\sigma) = \operatorname{Tr}\rho = 1$, the quantity $(\widetilde{Q}_\alpha - 1)/(\alpha - 1)$ is a difference quotient whose two-sided limit "should" be $\frac{d}{d\alpha}\widetilde{Q}_\alpha\big|_{\alpha=1} = D(\rho\|\sigma)$. The formal proof deliberately avoids establishing differentiability: a one-sided *eventual upper bound by a convergent majorant* suffices for the limit argument, and that follows from continuity of $\widetilde{D}_\alpha$ alone, via the two elementary scalar bounds.

6. Proof of the theorem

Fix $\rho_1, \rho_2, \sigma_1, \sigma_2$ and p. The proof splits exactly as the Lean source does.

Step 0 — degenerate weights. If $p = 0$, then $\bar\rho = \rho_2$, $\bar\sigma = \sigma_2$ (in Lean via `Mixable.mix_zero`, `Prob.lean:279`), and the right-hand side is $0 \cdot D(\rho_1\|\sigma_1) + 1 \cdot D(\rho_2\|\sigma_2) = D(\rho_2\|\sigma_2)$ — note this uses $0 \cdot \infty = 0$, so the case is genuine, not vacuous. Equality holds. If $p = 1$, symmetrically $\bar\rho = \rho_1$, $\bar\sigma = \sigma_1$ via the companion mixture identity `Mixable.mix_one` (`Prob.lean:284`, a new `@[simp]` lemma added by this contribution), and the right side is $D(\rho_1\|\sigma_1)$. Both degenerate cases now close by `simp` alone, the two `@[simp]` mixture lemmas discharging the mixture identity. Henceforth $0 < p < 1$ (Lean: `hp0'`, `hp1'`).

Step 1 — infinite right-hand side. Suppose $\operatorname{supp}\rho_1 \nsubseteq \operatorname{supp}\sigma_1$ (Lean: $\texttt{hker}_1$ fails). Then $D(\rho_1\|\sigma_1) = \infty$ by Definition 1 — in Lean via `qRelativeEnt_eq_top_iff.mpr` $\texttt{hker}_1$ (`Relative.lean:1795`), one of the two new finiteness characterizations — and since $p \neq 0$,

$p \cdot \infty = \infty$, so the right-hand side is ∞ and the inequality is trivial (`le_top`). The case $\operatorname{supp}\rho_2 \not\subseteq \operatorname{supp}\sigma_2$ is identical, using $1 - p \neq 0$. So we may assume both support conditions

$$\operatorname{supp}\rho_i \subseteq \operatorname{supp}\sigma_i \quad (i = 1, 2),$$

whence $D(\rho_1\|\sigma_1)$, $D(\rho_2\|\sigma_2) < \infty$; in Lean this finiteness is supplied where needed by the dual lemma `qRelativeEnt_ne_top_iff.mpr hker_i` (`Relative.lean:1789`).

Step 2 — setup for the main case. Write the mixtures as weighted sums over $\iota = \{1, 2\}$ (Lean: `Fin 2`) with weight vector $w = (p, 1 - p)$:

$$\bar\rho = \textstyle\sum_i w_i\rho_i, \qquad \bar\sigma = \textstyle\sum_i w_i\sigma_i$$

as matrix identities. In Lean this weighted-sum form is packaged, once and for all, as the lemma `mix_M_eq_weighted_sum` (`DPI.lean:1485`), which unfolds `Mixable.mix` into $\sum_i w_i(\cdot)_i$ with $w = (p, 1 - p)$. By Lemma C, $\operatorname{supp}\bar\rho \subseteq \operatorname{supp}\bar\sigma$; the weight nonnegativity, the weight sum $\sum_i w_i = 1$, and the per-component support conditions are all discharged inside the dedicated lemma `ker_mix_le` (`DPI.lean:1493`), so the theorem simply sets `hker_mix := ker_mix_le p hker`$_1$ `hker`$_2$. Apply Lemma E to each pair to obtain majorants u_1, u_2 with $u_i(\alpha) \to D(\rho_i\|\sigma_i)$ as $\alpha \to 1^+$ and, eventually, $(\widetilde{Q}_\alpha(\rho_i\|\sigma_i) - 1)/(\alpha - 1) \le u_i(\alpha)$.

Step 3 — pointwise bound for $\alpha > 1$ (Lean: `h_ev`). The binary case of Proposition A, specialized to a `Mixable` mixture, is packaged as the lemma `sandwichedTraceFunctional_mix_le` (`DPI.lean:1504`); `h_ev` invokes it as `sandwichedTraceFunctional_mix_le h`α `p hker`$_1$ `hker`$_2$. For $\alpha > 1$ in the eventual range of both majorants, using $\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma) > 0$ (positivity, with `hker_mix`) and Definition 3:

$$\widetilde{D}_\alpha(\bar\rho\|\bar\sigma) = \Big[\frac{\log\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma)}{\alpha - 1}\Big]^+,$$

and we bound the bracketed real number:

$$\begin{aligned}
\frac{\log\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma)}{\alpha-1} &\le \frac{\widetilde{Q}_\alpha(\bar\rho\|\bar\sigma) - 1}{\alpha - 1} && \text{[D(i): } \log x \le x - 1\text{]}\\
&\le \frac{p\,\widetilde{Q}_\alpha(\rho_1\|\sigma_1) + (1-p)\,\widetilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && \text{[Prop. A at } \iota = \{1,2\}\text{: } \texttt{sandwichedTraceFunctional_mix_le}\text{]}\\
&= p\,\frac{\widetilde{Q}_\alpha(\rho_1\|\sigma_1) - 1}{\alpha - 1} + (1-p)\,\frac{\widetilde{Q}_\alpha(\rho_2\|\sigma_2) - 1}{\alpha - 1} && \text{[since } p + (1-p) = 1\text{]}\\
&\le p\,u_1(\alpha) + (1-p)\,u_2(\alpha) && \text{[Lemma E].}
\end{aligned}$$

Hence, eventually as $\alpha \to 1^+$,

$$\widetilde{D}_\alpha(\bar\rho\|\bar\sigma) \;\le\; \big[\,p\,u_1(\alpha) + (1-p)\,u_2(\alpha)\,\big]^+.$$

Step 4 — pass to the limit $\alpha \to 1^+$. By Proposition B (applied to the mixture pair through `sandwichedRelRentropy_tendsto_qRelativeEnt`, `DPI.lean:1423`), the left side tends to $D(\bar\rho\|\bar\sigma)$ (Lean: `h_lhs`). By Lemma E and continuity of $t \mapsto [t]^+$, the right side tends to

$$\big[\,p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2)\,\big]^+ \;=\; p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2)$$

(Lean: `h_rhs`; the identification of the $[\,\cdot\,]^+$ of the real convex combination with the $[0, \infty]$-valued convex combination is the inline `have h_id` (`DPI.lean:1561`), which consumes the finiteness from Step 1 supplied by `qRelativeEnt_ne_top_iff`). Since the inequality of Step 3 holds eventually along the (nontrivial) filter of right-neighbourhoods of 1, the inequality passes to the limits (`le_of_tendsto_of_tendsto`):

$$D(\bar\rho\|\bar\sigma) \;\le\; p\,D(\rho_1\|\sigma_1) + (1-p)\,D(\rho_2\|\sigma_2). \qquad \blacksquare$$

7. Lean–paper dictionary

Paper object / step	Lean identifier	Location (commit `720c9fff`)
Umegaki D	`qRelativeEnt` / $\mathbf{D}$	`Relative.lean:1497`
Sandwiched $\widetilde{D}_\alpha$	`SandwichedRelRentropy` / `D̃_α`	`Relative.lean:1480`
Trace functional $\widetilde{Q}_\alpha$	`sandwichedTraceFunctional` / `Q̃_α`	`DPI.lean:66`
$\widetilde{D}_\alpha = [\log \widetilde{Q}_\alpha/(\alpha-1)]^+$	`sandwichedRelRentropy_eq_logTraceFunctional`	`DPI.lean:78`
Proposition A	`sandwichedTraceFunctional_jointly_convex`	`DPI.lean:568`
— variational formula	`traceFunctional_eq_iSup_f_alpha`, `f_alpha`, `H_hat`	`DPI.lean:434, 174, 180`
— concavity in σ	`HermitianMat.trace_conj_rpow_concave`	`LiebConcavity.lean:494`
Proposition B	`sandwichedRelRentropy.continuousOn`	`Relative.lean:1769`
— packaged as $\alpha \to 1^+$ convergence	`sandwichedRelRentropy_tendsto_qRelativeEnt`	`DPI.lean:1423`
Lemma C	`HermitianMat.ker_weighted_sum_le`	`DPI.lean:545`
D(i)	`Real.log_le_sub_one_of_pos`	Mathlib
D(ii)	`Real.abs_exp_sub_one_sub_id_le`	Mathlib
$\widetilde{Q}_\alpha > 0$	`sandwichedTraceFunctional_pos`	`DPI.lean:100`
Lemma E	`sandwichedTraceFunctional_sub_one_div_eventually_le`	`DPI.lean:1432`
$\mathbf{D} \neq \top \iff$ supp-inclusion (Step 1 finiteness)	`qRelativeEnt_ne_top_iff`	`Relative.lean:1789`
$\mathbf{D} = \top \iff$ supp-inclusion fails (Step 1 ∞-branch)	`qRelativeEnt_eq_top_iff`	`Relative.lean:1795`
Mixture as $\sum_i w_i(\cdot)_i$ (Step 2 matrix identity)	`mix_M_eq_weighted_sum`	`DPI.lean:1485`
$\operatorname{supp}\bar{\rho} \subseteq \operatorname{supp}\bar{\sigma}$ (Step 2, `hker_mix`)	`ker_mix_le`	`DPI.lean:1493`
Binary $\widetilde{Q}_\alpha$ convexity (Step 3)	`sandwichedTraceFunctional_mix_le`	`DPI.lean:1504`
ofReal of the convex combination (Step 4 identity)	inline `have h_id`	`DPI.lean:1561`
$p = 0$ / $p = 1$ degenerate mixtures (Step 0)	`Mixable.mix_zero` / `Mixable.mix_one`	`Prob.lean:279, 284`
Steps 0–4	`qRelativeEnt_joint_convexity`	`DPI.lean:1522`
Step 3	`h_ev` (calling `sandwichedTraceFunctional_mix_le`)	in theorem body, `DPI.lean:1522`
Step 4	`h_lhs`, `h_rhs`, `le_of_tendsto_of_tendsto`	in theorem body, `DPI.lean:1522`

8. Remarks for the reviewer

- **What is new.** Lemma E and the theorem itself, together with the supporting lemmas that carry the individual proof steps — the public `sandwichedRelRentropy_tendsto_qRelativeEnt` and the private `mix_M_eq_weighted_sum`, `ker_mix_le`, `sandwichedTraceFunctional_mix_le` — account for the +190 lines added to `DPI.lean` (§5–§6). In `Relative.lean` the previously

@[sorryful] statement was removed and replaced by the two finiteness characterizations qRelativeEnt_ne_top_iff / qRelativeEnt_eq_top_iff (Relative.lean:1789, 1795; net +11 / −15), and Prob.lean gains one @[simp] lemma, Mixable.mix_one (Prob.lean:284, +5). Propositions A and B and all deeper inputs (Lieb–Ando trace theorems) were already in the library. The theorem was placed in DPI.lean rather than Relative.lean because the $\widetilde{Q}_\alpha$ machinery lives there and importing it from Relative.lean would create a cycle.

- **No differentiability is needed.** The classical way to see $\widetilde{D}_\alpha \to D$ and to transfer convexity is to differentiate $\alpha \mapsto \widetilde{Q}_\alpha$ at $\alpha = 1$; the formal proof replaces this by the one-sided majorant of Lemma E, which needs only continuity (Proposition B) plus the scalar bounds D. This is the main proof-engineering choice.
- **The statement is the binary form.** Finite mixtures of n states follow by induction in the usual way, but only the n-ary form of Proposition A (arbitrary finite Fintype ι) is formalized; the **D**-level statement is for two pairs, matching the pre-existing sorryful statement it closes.
- **Axioms.** The proof is sorry-free; #print axioms qRelativeEnt_joint_convexity reports only propext, Classical.choice, Quot.sound (recorded in the commit message of 720c9fff).